

Abstract algebra – 1

Finite groups: counting

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Theoretical minimum. Everything used below is proved here, so no course in algebra is presupposed.

Definitions. A *group* is a set G with an associative operation, a neutral element e and inverses. A *subgroup* $H \leq G$ is a subset closed under the operation and under inversion. The *order* $\text{ord}(x)$ of an element is the least $k > 0$ with $x^k = e$; in a finite group it always exists, because the powers of x repeat. The group is *cyclic* if $G = \langle x \rangle = \{e, x, \dots, x^{n-1}\}$ for some x . The *centre* is $Z(G) = \{a : ax = xa \text{ for all } x\}$, the *centraliser* of x is $C(x) = \{a : ax = xa\}$, and the *index* $[G : H]$ is the number of left cosets xH .

Theorem 1 (Lagrange). The left cosets xH partition G and all have $|H|$ elements, hence $|G| = [G : H] \cdot |H|$ and $|H|$ divides $|G|$. *Proof:* $x \in xH$; if $xH \cap yH \neq \emptyset$ then $xh_1 = yh_2$, so $x = yh_2h_1^{-1}$ and $xH \subseteq yH$, and by symmetry $xH = yH$; the map $h \mapsto xh$ is a bijection $H \rightarrow xH$. *Corollaries:* $\text{ord}(x)$ divides $|G|$ (take $H = \langle x \rangle$), $x^{|G|} = e$ for every x , a group of prime order is cyclic, and for $G = (\mathbb{Z}/n\mathbb{Z})^*$ one gets Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem 2 (Cauchy). If a prime p divides $|G|$, then G has an element of order p . *Proof (McKay):* let $X = \{(x_1, \dots, x_p) \in G^p : x_1x_2 \cdots x_p = e\}$. The first $p-1$ coordinates are arbitrary and the last one is determined, so $|X| = |G|^{p-1}$ is divisible by p . The cyclic shift $(x_1, \dots, x_p) \mapsto (x_2, \dots, x_p, x_1)$ maps X to itself, because $x_1(x_2 \cdots x_p) = e$ implies $(x_2 \cdots x_p)x_1 = e$. Its orbits have length 1 or p , so the number of fixed points is divisible by p ; the fixed points are the tuples $(x, \dots, x)$ with $x^p = e$, and $(e, \dots, e)$ is one of them, so there are at least p of them.

Theorem 3 (the involution count). In a finite group $\#\{x : x^2 = e\} \equiv |G| \pmod{2}$. *Proof:* the elements with $x^2 \neq e$ split into pairs $\{x, x^{-1}\}$. In

particular a group of even order contains an element of order 2, which is Cauchy's theorem for $p = 2$ obtained by pure counting.

Theorem 4 (the class equation). Conjugacy $x \sim gxg^{-1}$ is an equivalence relation, the class of x has $[G : C(x)]$ elements, and classes of size 1 are exactly the central elements. Hence $|G| = |Z(G)| + \sum [G : C(x_i)]$, the sum being over representatives of the classes of size > 1 . *Corollary:* if $|G| = p^k$ then every term of the sum is divisible by p , so p divides $|Z(G)|$ and the centre is non-trivial.

Fact 1 (centre and commutativity). If $G/Z(G)$ is cyclic then G is abelian: writing $x = z_1 g^i$, $y = z_2 g^j$ with central z_1, z_2 gives $xy = yx$. Consequently a group of order p^2 is abelian.

Theorem 5 (finite abelian groups). Every finite abelian group is a direct sum of cyclic groups $\mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$ with $d_1 \mid d_2 \mid \cdots \mid d_k$; the number d_k is the exponent, that is the least m with $x^m = e$ for all x . A finite abelian group whose exponent equals its order is cyclic; a group in which $x^2 = e$ for all x is abelian and isomorphic to $(\mathbb{Z}/2)^k$, so its order is a power of two.

1. Prove Lagrange's theorem and deduce: $\text{ord}(x) \mid |G|$; $x^{|G|} = e$; a group of prime order is cyclic; Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$; and Fermat's little theorem as the case $n = p$. Then show that the converse of Lagrange fails: exhibit a group of order 12 with no subgroup of order 6.
2. Prove that $\#\{x \in G : x^2 = e\}$ and $|G|$ have the same parity, and deduce that a group of even order contains an involution. Then reconstruct McKay's proof of Cauchy's theorem for a general prime p : describe the set on which the cyclic shift acts, compute its cardinality, and explain why the number of fixed points is divisible by p .
3. Suppose that a finite group has exactly n elements of order p , where p is a prime. Prove that either $n = 0$ or p divides $n + 1$. (Putnam 2007,

A5)

4. Let G be a finite group of order $n > 2$ and let p be the smallest prime divisor of n . Prove that if G has a unique subgroup H with p elements, then H is contained in the centre of G . (Румыния, 2006)
5. Is there a finite abelian group G such that the product of the orders of all its elements is 2^{2009} ? (Putnam 2009, A5)